

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

©:Zac Kienzle[†]

[†] University of Queensland, Brisbane, Australia

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Part I

Mathematical Foundations

1 Measure and Integration

2 Probability

3 Convergence

4 Conditioning

4.1 Conditional Expectation

In elementary probability, we define the conditional expectation after defining conditional probability. In advanced probability, it is more convenient to reverse the order, as we often want to condition on σ -algebras. In what follows, $(\Omega, \mathcal{H}, \mathbb{P})$ is a probability space, X a random variable, and $\mathcal{F}$ a sub- σ -algebra of $\mathcal{H}$; that is a σ -algebra whose every member (i.e., set) belongs to $\mathcal{H}$ as well. Recall the notation $V \in \mathcal{F}_+$ to mean that $V \geq 0$ and is $\mathcal{F}$ -measurable.

Definition 4.1: Conditional Expectation

The conditional expectation of X given $\mathcal{F}$, denoted $\mathbb{E}_{\mathcal{F}}X$, is defined in two steps:

- (1) For $X \geq 0$, $\mathbb{E}_{\mathcal{F}}X$ is defined as any random variable $\hat{X}$ with
 - (1) $\hat{X} \in \mathcal{F}_+$.
 - (2) $\mathbb{E}VX = \mathbb{E}V\hat{X}$ for every $V \in \mathcal{F}_+$ (*projection property*).
- (2) For arbitrary X , if $\mathbb{E}X$ exists (possibly $\pm\infty$), $\mathbb{E}_{\mathcal{F}}X := \mathbb{E}_{\mathcal{F}}X^+ - \mathbb{E}_{\mathcal{F}}X^-$. Otherwise, the conditional expectation is left undefined.

Note that the conditional expectation is defined *almost surely*.

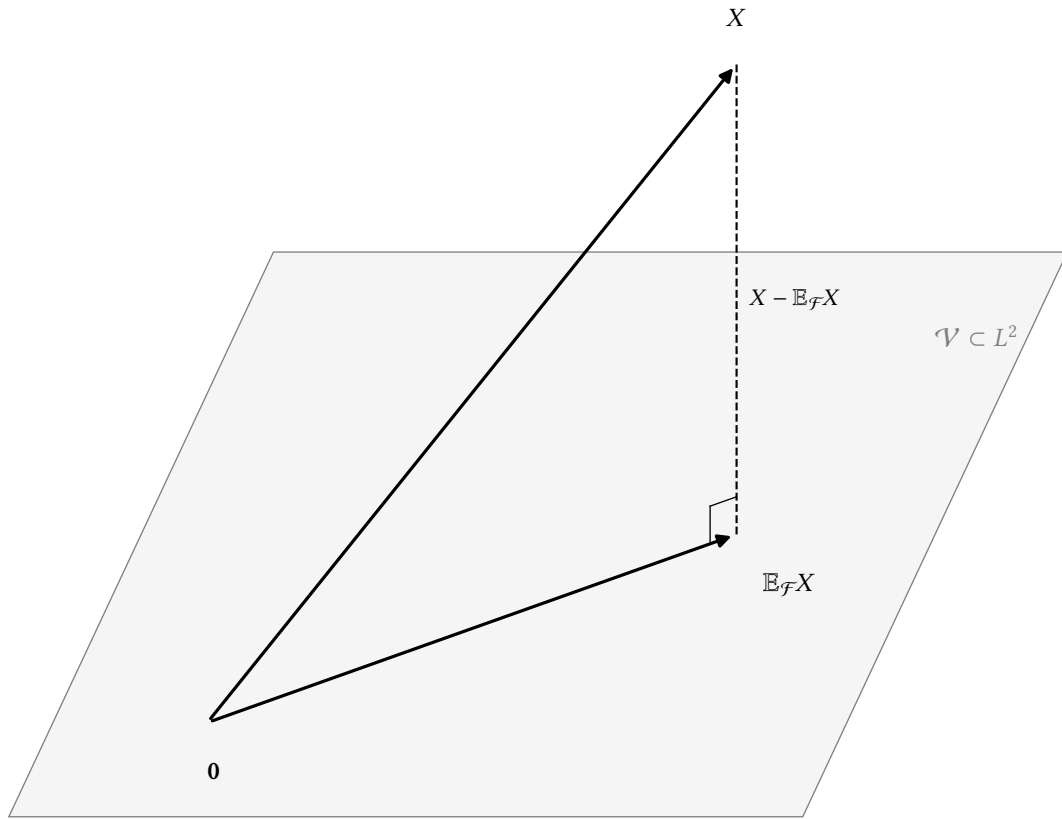


Figure 1: Geometric interpretation of Conditional Expectation $\mathbb{E}_{\mathcal{F}}X$ as an orthogonal projection.

Theorem 4.1: Existence and Uniqueness of the Conditional Expectation

Let $X \geq 0$ be a random variable and let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. Then, $\mathbb{E}_{\mathcal{F}}X$ exists and is unique up to equivalence.

Theorem 4.2: Additional Properties of the Conditional Expectation

Let $\mathcal{F}$ and $\mathcal{G}$ be sub- σ -algebras of $\mathcal{H}$. Let W and X be random variables such that $\mathbb{E}X$ and $\mathbb{E}WX$ exist. Then, the following hold:

- (1) (Taking out what is known): If $W \in \mathcal{F}$, then $\mathbb{E}_{\mathcal{F}}WX = W\mathbb{E}_{\mathcal{F}}X$.
- (2) (Repeated conditioning): If $\mathcal{F} \subseteq \mathcal{G}$, then

$$\mathbb{E}_{\mathcal{F}}\mathbb{E}_{\mathcal{G}}X = \mathbb{E}_{\mathcal{G}}\mathbb{E}_{\mathcal{F}}X = \mathbb{E}_{\mathcal{F}}X. \quad (4.1)$$

There is an amusing way to remember repeated conditioning: the σ -algebras $\mathcal{F}$ and $\mathcal{G}$ represent the information about the measurement on a random experiment by a fool and a genius. The fool has no use for the genius' information,

$$\mathbb{E}_{\mathcal{F}}\mathbb{E}_{\mathcal{G}}X = \mathbb{E}_{\mathcal{F}}X,$$

and, sadly, the genius cannot further improve on the poor information of the fool,

$$\mathbb{E}_{\mathcal{G}}\mathbb{E}_{\mathcal{F}}X = \mathbb{E}_{\mathcal{F}}X.$$

In many applications of conditioning, we wish to take the expectation of a random variable X conditional on one or more random variables or, in general, on a stochastic process $Y := \{Y_t, t \in \mathbb{T}\}$, in which case $\mathcal{F} = \sigma(Y) = \sigma\{Y_t, t \in \mathbb{T}\}$. In this case we often write

$$\mathbb{E}_{\mathcal{F}}X = \mathbb{E}[X \mid Y_t, t \in \mathbb{T}] = \mathbb{E}[X \mid Y].$$

When conditioning on a random variable Y , we can use the fact (see ??) that any random variable in $\sigma(Y)$, in particular $\mathbb{E}[X \mid Y]$, must be a deterministic measurable function of Y ; that is, $\mathbb{E}[X \mid Y] = h(Y)$ for some measurable function h . This is in accordance with our use of conditional expectations in elementary probability: we first determine $h(y) := \mathbb{E}[X \mid Y = y]$ and then use $\mathbb{E}[X \mid Y] = h(Y)$ in computations.

Theorem 4.3: Conditional Expectation in L^2

For $X \in L^2$, the conditional expectation $\mathbb{E}_{\mathcal{F}}X$ is the orthogonal projection of X onto the subspace $\mathcal{V}$ of $\mathcal{F}$ -measurable random variables in L^2 .

4.2 Conditional Probability and Distribution

Definition 4.2: Conditional Probability for Events

Let A and B be two events with $\mathbb{P}(B) > 0$. The *conditional probability* that A occurs given that B occurs is denoted by

$$\mathbb{P}(A | B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (4.2)$$

Theorem 4.4: Properties Involving Conditional Probabilities

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $A, A_1, A_2, \dots, A_n, B_1, B_2, \dots$ events such that $\mathbb{P}(A_1 \cap \dots \cap A_n) > 0$ and $\{B_i\}$ form a partition of Ω , with $\mathbb{P}(B_i) > 0$ for all i . Then, the following hold:

(1) Law of total probability: $\mathbb{P}(A) = \sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i)$.

(2) Bayes' rule: $\mathbb{P}(B_j | A) = \frac{\mathbb{P}(B_j) \mathbb{P}(A | B_j)}{\sum_i \mathbb{P}(B_i) \mathbb{P}(A | B_i)}$.

(3) Chain or product rule:

$$\mathbb{P}(\cap_{i=1}^n A_i) = \mathbb{P}(A_1) \mathbb{P}(A_2 | A_1) \cdots \mathbb{P}(A_n | A_1 \cap \dots \cap A_{n-1}).$$

Definition 4.3: Conditional Distribution

Let Y be a random variable on the probability space $(\Omega, \mathcal{H}, \mathbb{P})$ taking values in the measurable space $(E, \mathcal{E})$. Let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. The *conditional distribution* of Y given $\mathcal{F}$ is any probability transition kernel L from $(\Omega, \mathcal{F})$ to $(E, \mathcal{E})$ such that

$$\mathbb{P}_{\mathcal{F}}(Y \in B) = L(\cdot, B), \quad B \in \mathcal{E}. \quad (4.3)$$

Theorem 4.5: Existence of the Conditional Distribution

For every standard measurable space $(E, \mathcal{E})$ there exists a version of the conditional distribution of Y given $\mathcal{F}$.

Theorem 4.6: Regular Version of a Conditional Probability

Let $(\Omega, \mathcal{H})$ be a standard measurable space and $\mathcal{F}$ a sub- σ -algebra of $\mathcal{H}$. Then, there exists a regular version of the conditional probability $\mathbb{P}_{\mathcal{F}}$.

$$\pi(dx, dy) = \mu(dx)K(x, dy). \quad (4.4)$$

Theorem 4.7: Conditional Distribution

If π is of the form eq. (4.4), then the kernel L defined by

$$L(\omega, B) := K(X(\omega), B), \quad \omega \in \Omega, B \in \mathcal{E} \quad (4.5)$$

is a version of the conditional distribution of Y given X .

Theorem 4.8: Disintegration

Let π be a probability measure on the product space $(D \times E, \mathcal{D} \otimes \mathcal{E})$, where $(E, \mathcal{E})$ is a standard measurable space. Then, there exists a probability measure μ on $(D, \mathcal{D})$ and a transition kernel K from $(D, \mathcal{D})$ to $(E, \mathcal{E})$ such that

$$\pi(dx, dy) = \mu(dx)K(x, dy), \quad x \in D, y \in E. \quad (4.6)$$

Definition 4.4: Conditional Independence

Let $\mathcal{F}, \mathcal{F}_1, \dots, \mathcal{F}_n$ be sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be *conditionally independent* given $\mathcal{F}$ if

$$\mathbb{E}_{\mathcal{F}} V_1 \dots V_n = \mathbb{E}_{\mathcal{F}} V_1 \dots \mathbb{E}_{\mathcal{F}} V_n \quad (4.7)$$

for all positive random variables $V_1, \dots, V_n$ in $\mathcal{F}_1, \dots, \mathcal{F}_n$, respectively.

Proposition 4.1: Conditional Independence

$\mathcal{F}_1$ and $\mathcal{F}_2$ are conditionally independent given $\mathcal{F}$ if and only if

$$\mathbb{E}_{\mathcal{F} \vee \mathcal{F}_1} V_2 = \mathbb{E}_{\mathcal{F}} V_2 \quad \text{for all positive } V_2 \in \mathcal{F}_2. \quad (4.8)$$

$$\mathbb{P}(B_0 \times \dots \times B_n \times E_{n+1} \times \dots) = \pi_n(B_0 \times \dots \times B_n), \quad B_i \in \mathcal{E}_i, i = 0, \dots, n. \quad (4.9)$$

4.3 Existence of Probability Spaces

Theorem 4.9: Ionescu–Tulcea’s Existence Theorem

There exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ such that eq. (4.9) holds.

Theorem 4.10: Kolmogorov’s Extension Theorem

Let $(E, \mathcal{E})$ be a standard measurable space and K an index set. For each finite subset J of K , let π_J be a probability measure on the product space $(E^J, \mathcal{E}^J)$. If the (π_J) are consistent in the sense that

$$\pi_I = \pi_J \circ \text{proj}_{JI}^{-1}, \quad I \subseteq J, \quad |J| < \infty,$$

then there exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ and a stochastic process X taking values in $(E, \mathcal{E})^K$ such that, with $X_J := (X_j)_{j \in J}$:

$$\mathbb{P}(X_J \in A) = \pi_J(A), \quad A \in \mathcal{E}^J, \quad |J| < \infty.$$

Definition 4.5: Hölder Continuity

A function $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ is said to be *Hölder continuous* of order $\alpha > 0$ on $B \subseteq \mathbb{R}_+$ if there is a constant k such that

$$|f(t) - f(s)| \leq k|t - s|^\alpha, \quad s, t \in B.$$

Theorem 4.11: Hölder Continuous Modification

Let $X := (X_t, t \in [0, 1])$ be a stochastic process with state space $\mathbb{R}$. If there exist constants c, p, q in $(0, \infty)$ such that

$$\mathbb{E}|X_t - X_s|^p \leq c|t - s|^{1+q}, \quad s, t \in [0, 1], \quad (4.10)$$

then for every $\alpha \in (0, q/p)$ there is a modification $\tilde{X}$ of X that is almost surely Hölder continuous of order α on $[0, 1]$.

4.4 Markov Property

One of the most studied stochastic processes is the Markov process. In a way it can be viewed as the probabilistic analogue of a difference or differential equation, in the sense that, conditional on the past history, future increments depend only on the present state. The concepts of conditional probability and expectation are thus essential to the analysis.

Throughout this section, $\mathbb{T}$ is some subset of $\mathbb{R}$ and $X := (X_t, t \in \mathbb{T})$ is a stochastic process on some probability space $(\Omega, \mathcal{H}, \mathbb{P})$, taking values in some measurable space $(E, \mathcal{E})$.

Definition 4.6: Adaptedness

A stochastic process $X := (X_t, t \in \mathbb{T})$ taking values in some measurable space $(E, \mathcal{E})$ is said to be *adapted* to a filtration $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$ if for all $t \in \mathbb{T}$, X_t is $\mathcal{F}_t/\mathcal{E}$ -measurable.

In what follows, we assume that the process X is adapted to some filtration $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$ – for example, the natural filtration of X , in which case $\mathcal{F}_t := \sigma(X_u, u \leq t, u \in \mathbb{T})$. Let $\mathcal{G}_t := \sigma(X_u, u \geq t, u \in \mathbb{T})$ be the *future* σ -algebra of X for time t .

Definition 4.7: Markov Property

X is said to be *Markovian* relative to $\mathcal{F}$ if for every t , the past $\mathcal{F}_t$ and the future $\mathcal{G}_t$ are conditionally independent, given the present state X_t .

Definition 4.8: Markovian Transition Function

A *Markovian transition function* on $(E, \mathcal{E})$ is a family $(P_{t,u}, t, u \in \mathbb{T}, t < u)$ of probability transition kernels on $(E, \mathcal{E})$ that satisfy the *Chapman–Kolmogorov equations*:

$$P_{s,t}P_{t,u} = P_{s,u}, \quad s < t \leq u. \quad (4.11)$$

A Markovian process X is said to admit $(P_{t,u})$ as a transition function if

$$\mathbb{E}[f(X_u) | X_t] = (P_{t,u}f)(X_t), \quad t < u, f \in \mathcal{E}_+.$$

For $f = \mathbb{1}_A$ with $A \in \mathcal{E}$, this means

$$\mathbb{P}(X_u \in A \mid X_t = x) = P_{t,u}(x, A),$$

giving an intuitive meaning to the transition functions.

Define $P_t := P_{0,t}$. If $P_{t,u} = P_{u-t}$ for all $t \leq u$, the associated Markovian process X is said to be *time-homogeneous* and P_t satisfies the *semi-group property*

$$P_{t+s} = P_s P_t, \quad s, t \geq 0.$$

Markov processes can be classified according to the nature (discrete or continuous) of their state space E and time set $\mathbb{T}$. We call a Markovian process a *Markov chain*¹ if $\mathbb{T}$ is discrete. The Markov process *par excellence* in continuous time is the Lévy process; this includes the Wiener, Poisson, compound Poisson, and pure-jump Lévy processes.

Theorem 4.12: Generator of a Markov Jump Process

The infinitesimal generator of a Markov jump process with Lévy kernel L is given by

$$(Gf)(x) = \int_{\mathbb{R}^d} L(x, dy)[f(y) - f(x)], \quad f \in \mathcal{B}^d(\text{bounded}). \quad (4.12)$$

Proposition 4.2: Poisson Integrals

Let N be a Poisson random measure on $\mathbb{R}_+ \times E$ with mean measure μ satisfying $\mu(\{0\} \times E) = 0$. Let $R := (R(t, z), t \geq 0, z \in E)$ be a positive stochastic process in $\mathcal{F}^P \otimes \mathcal{E}$. Then,

$$\mathbb{E}NR = \mathbb{E}\mu R. \quad (4.13)$$

¹There is no consensus on this nomenclature. Some call a Markovian process a Markov chain if E is discrete.

5 Martingales

6 Wiener and Brownian Motion Processes

7 Itô Calculus

Part II

Levy Processes

Part III

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Part VI

Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate

simulated option price values, which are then compared against outcomes derived from analytical formulas.

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17 Quasi Monte Carlo

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